

# E-Commerce Return Rate Reduction Analysis

## Project Report

### Abstract:

Product returns are a significant cost driver for e-commerce businesses, affecting profitability, logistics, and sustainability targets. This project analyzes a 5,000-order e-commerce return and sustainability dataset to understand why customers return products and how return behaviour varies across product category, shipping method, payment method, and customer geography. The dataset was cleaned and explored in Python, queried and aggregated using SQL, and visualized through an interactive Power BI dashboard. The overall return rate was found to be approximately 29%, with Clothing showing the highest category-level return rate (37.4%) and Defective items and Changed Mind being the two leading return reasons. The resulting dashboard highlights high-risk products, locations, and categories to support data-driven decisions aimed at reducing avoidable returns.

### Introduction:

Returns are an unavoidable part of online retail, but a high or poorly understood return rate erodes margins and increases packaging waste and carbon emissions. The objective of this project was to identify the key drivers behind product returns and quantify how return percentage, profit loss, and environmental impact (CO2 emissions and packaging waste) vary across product categories, suppliers, shipping methods, payment methods, and customer locations. The end goal was an interactive Return Risk dashboard that a business team could use to drill down into high-risk products and regions and prioritize corrective action.

### Tools Used:

- Python (Pandas, NumPy, Jupyter Notebook) — data loading, cleaning, type conversion, missing value and duplicate checks, and descriptive/category-wise statistical analysis.
- SQL (MySQL) — schema design with constraints, bulk data loading, and aggregate queries for return rate by category, product, location, shipping method, payment method, and profit/loss impact.
- Power BI — building an interactive, multi-page dashboard with KPI cards, trend lines, bar and pie charts, a treemap, and drill-through filters for return-risk exploration.

### Steps Involved in Building the Project:

- Data Cleaning (Python): Loaded the raw returns\_sustainability\_dataset.csv, inspected shape, data types, and summary statistics, converted Order\_Date to datetime, and verified there were no duplicate or missing records across the 5,000 orders.
- Exploratory Analysis (Python): Computed the overall return rate, and category-wise Total Orders / Returned Orders / Return Rate, along with distribution checks on price, discount, shipping method, payment method, and sustainability metrics (CO2 emissions, packaging waste, CO2 saved).
- Database & SQL Analysis: Created a structured MySQL table with appropriate data types and check constraints, bulk-loaded the dataset, and wrote aggregate queries covering: overall return rate; return rate by category, product, shipping method, and payment method; top 10 high-risk products and locations; return reason breakdown; and profit/loss impact of returns by category and return status.
- Dashboard Design (Power BI): Built a three-page interactive dashboard – (1) an executive overview with total orders, return rate, return cost, monthly trend, and return-reason and category breakdowns; (2) a return-risk drill-down view with a per-product table, category and age-group return rates; and (3) a product-level detail page with order value, return cost by reason, and a payment-category treemap – all connected through slicers and drill-through filters.
- Insight Generation: Consolidated the SQL and Power BI outputs to flag high-risk product categories (Clothing), leading return reasons (Defective, Changed Mind), and locations with disproportionately high return rates for follow-up action.

### Key Findings:

| Metric                       | Value                                   |
|------------------------------|-----------------------------------------|
| Overall Return Rate          | ~29%                                    |
| Highest Return Rate Category | Clothing (37.4%)                        |
| Lowest Return Rate Category  | Books (22.6%)                           |
| Top Return Reasons           | Defective (26.3%), Changed Mind (26.1%) |
| Total Return Cost            | ~2.9 Lakh (290K)                        |

## **Conclusion:**

The analysis shows that returns are not evenly distributed across the business: Clothing drives the highest return rate, and product defects and changed-mind decisions together account for over half of all returns. Shipping method had only a marginal effect on return likelihood, while a small set of products and locations contribute disproportionately to return volume and cost. Combining SQL-based aggregation with an interactive Power BI dashboard makes these patterns easy to monitor on an ongoing basis. Going forward, the same feature set (category, price, discount, shipping and payment method) can be extended into a logistic regression model to score each order's return probability, enabling the business to proactively target quality checks and communication for the highest-risk product categories, such as Clothing, before returns occur.